Project Closeout Report

Name:
Fluid Swaps – Trade CNTs against BTC, ETH or MATIC

URL:
https://milestones.projectcatalyst.io/projects/1300102

Your Project Number:
1300102

Name of project manager:
Matteo Coppola

Date project started:
March 2025

Date project completed:
February 2026

Challenge KPIs:

- Cross-chain interoperability: enable Cardano Native Tokens (CNTs) to be traded against assets on other blockchains in a trustless, decentralized manner
- Atomic swap execution: users can complete swaps between Cardano and Bitcoin/Ethereum without relying on centralized exchanges or custodians
- Open source delivery: all protocol components are publicly available and independently verifiable

List of project KPIs:

- Clear Project Scope and Objectives: Established and communicated clear scope from the beginning — decentralized cross-chain atomic swaps between Cardano, Bitcoin, and Ethereum using Hash Time-Locked Contracts (HTLCs)
- Smart contract development, auditing and release: Full development lifecycle from design to mainnet deployment, including a formal third-party security audit of all Cardano validators and cross-chain signature logic
- Cross-chain compatibility: Support for Bitcoin (BTC / Runes), Ethereum (ETH / ERC20), and Cardano Native Tokens through a unified HTLC mechanism
- Frontend development and deployment: A user-facing marketplace allowing users to create offers, accept offers, and execute atomic swaps end-to-end, with integrated wallet support across 10+ Cardano wallets, 5 Bitcoin wallets, and 2 Ethereum wallets
- Guideline and Instruction Documentation: Public user documentation and technical smart contract documentation released alongside the platform

Key achievements:

- First atomic swap between Cardano and Bitcoin on testnet successfully executed and announced

- First atomic swap on Cardano mainnet completed with real users
- Smart contracts written in Aiken (Plutus V3), with full support for both Bitcoin (secp256k1 + double SHA-256) and Ethereum (secp256k1 + Keccak-256) signature formats within the same on-chain validator
- Security audit completed and published (FluidSwaps Protocol Smart Contract Audit Report v3, Feb2026)
- Fully open source codebase under MIT license
- Live mainnet platform accessible at http://fluid-swap.fluidtokens.com/ with a public marketplace and dashboard
- Community-verified swaps, with independent users publicly sharing their swap experiences

Key learnings:

- Cross-chain atomic swaps require careful handling of blockchain-specific message signing conventions — Bitcoin and Ethereum use the same elliptic curve (secp256k1) but different hashing and magic byte prefixes, and supporting both within a single Cardano validator required precise low-level implementation
- Plutus V3 and Aiken provide the necessary cryptographic primitives (ECDSA secp256k1 verification, Keccak-256) that make truly trustless cross-chain swaps from the Cardano side feasible
- Asymmetric deadlines (shorter for the Cardano side, longer for the Bitcoin/Ethereum side) are essential to protect both parties given the difference in block confirmation times across chains
- User experience for multi-wallet flows (requiring two wallets from different chains simultaneously) requires clear UX design and thorough documentation to avoid user confusion during the swap lifecycle

Next steps for the service:

- Expand CNT marketplace listings to support a wider range of Cardano Native Tokens
- Explore Runes (Bitcoin-native tokens) support for CNT ↔ Runes atomic swaps
- Integration with additional Bitcoin wallet providers
- Explore trustless BTC bridging flows to Cardano DeFi, complementing the Aquarium Babel Fees infrastructure

Final thoughts/comments:

Fluid Swaps demonstrates that fully trustless, non-custodial cross-chain trading between Cardano and Bitcoin/Ethereum is technically achievable today, without bridges, wrapped tokens, or centralized relayers. The HTLC mechanism, implemented natively in Aiken on Cardano with cross-chain signature verification, provides a security-first foundation for cross-chain DeFi on Cardano. With mainnet live and real swaps confirmed by independent users, the project has delivered on all core technical commitments.

Links to other relevant project sources or documents:

Public documentation:

https://github.com/FluidTokens/ft-cardano-fluidswaps-sc/blob/main/FluidSwap_Docs.md

GitHub Repository (smart contracts):

https://github.com/FluidTokens/ft-cardano-fluidswaps-sc

Public README:

https://github.com/FluidTokens/ft-cardano-fluidswaps-sc/blob/main/README.md

First atomic swap on testnet (announcement):

https://x.com/FluidTokens/status/2000981697672696122

First atomic swap on mainnet (announcement):

https://x.com/FluidTokens/status/2037244182130020532

Video demo: https://youtu.be/gLg3HIOE0LM

Live video demo: https://x.com/elraulito/status/2001324294908035477

Closeout video: https://youtu.be/hrUwx3pKQbM

User-tested swap: https://x.com/silversoul8668/status/2024473201238299061

Live mainnet platform: http://fluid-swap.fluidtokens.com/